

Week 5 — Session 4

Course: 10211CS225 — Problem Solving and Testing using Java

Unit II: Algorithmic Thinking & Competitive Problem Patterns

Topic: Advanced String Techniques

Language: Java

Focus: Use sliding-window and bijective mapping techniques for more advanced string problems.

Task 1: Longest Substring Without Repeating Characters

Practice: <https://leetcode.com/problems/longest-substring-without-repeating-characters/>

Complete Java Solution

```
import java.util.*;

public class Main {
    static int longest(String s){
        Map<Character,Integer> last=new HashMap<>();
        int left=0,best=0;

        for(int right=0;right<s.length();right++){
            char c=s.charAt(right);
            if(last.containsKey(c))
                left=Math.max(left,last.get(c)+1);
            last.put(c,right);
            best=Math.max(best,right-left+1);
        }
        return best;
    }

    public static void main(String[] args){
        Scanner sc=new Scanner(System.in);
        System.out.println(longest(sc.nextLine()));
        sc.close();
    }
}
```

Task 2: Find and Replace Pattern

Practice: <https://leetcode.com/problems/find-and-replace-pattern/>

Complete Java Solution

```
import java.util.*;

public class Main {
    static boolean matches(String word,String pattern){
        Map<Character,Character> pToW=new HashMap<>();
        Map<Character,Character> wToP=new HashMap<>();

        for(int i=0;i<word.length();i++){
            char w=word.charAt(i), p=pattern.charAt(i);

            if(pToW.containsKey(p) && pToW.get(p)!=w) return false;
            if(wToP.containsKey(w) && wToP.get(w)!=p) return false;

            pToW.put(p,w);
            wToP.put(w,p);
        }
        return true;
    }

    public static void main(String[] args){
        Scanner sc=new Scanner(System.in);
        int n=sc.nextInt();
        String[] words=new String[n];
        for(int i=0;i<n;i++) words[i]=sc.next();
        String pattern=sc.next();

        for(String w:words)
            if(matches(w,pattern)) System.out.print(w+" ");
        System.out.println();
        sc.close();
    }
}
```

Programs are standalone Java solutions. Java 17+ is recommended.